

PROGRAMME: HIGHER NATIONAL DIPLOMA IN ACCOUNTANCY

Code: ACC 327 /BAM 321

Course: MANAGEMENT INFORMATION SYSTEM II

Goal: This course is designed to acquaint the student with further application and use of Computers in industry and commerce and with further developments in information system.

GENERAL OBJECTIVES:

On completion of this course, the student should be able to:

- 1.0 Understand Information System.
- 2.0 Understand Transaction Processing System (TPS)
- 3.0 Understand Management Information System.
- 4.0 Understand Decision Tables.
- 5.0 Understand Project Planning and Scheduling

1.0 The Concept of Management Information System (MIS)

1.1 DEFINITION OF MIS To the managers, Management Information System is an implementation of the organizational systems and procedures.

To a programmer it is nothing but files structures and file processing. However, it involves much more complexity. The three components of MIS provide a more complete and focused definition, where System suggests integration and holistic view, Information stands for processed data, and Management is the ultimate user, the decision makers.

Management Information System can thus be analyzed as follows: Management covers the planning, control, and administration of the operations of a concern. The top management handles planning; the middle management concentrates on controlling; and the lower management is concerned with actual administration.

Information, in MIS, means the processed data that helps the management in planning, controlling and operations.

Data means all the facts arising out of the operations of the concern. Data is processed i.e. recorded, summarized, compared and finally presented to the management in the form of MIS report. System Data is processed into information with the help of a system. A system is made up of inputs, processing, output and feedback or control. Thus MIS means a system for processing data in order to give proper information to the management for performing its functions. Definition Management Information System or 'MIS' is a planned system of collecting, storing, and disseminating data in the form of information needed to carry out the functions of management.

Management Information Systems (MIS), referred to as Information Management and Systems, is the discipline covering the application of people, technologies, and procedures collectively called information systems, to solving business problems. MIS is a planned system of collecting, storing and disseminating data in the form of information needed to carry out the functions of management.

Objectives of MIS The goals of an MIS are to implement the organizational structure and dynamics of the enterprise for the purpose of managing the organization in a better way and capturing the potential of the information system for competitive advantage.

Followings are the basic objectives of an MIS:

- i. **Capturing Data:** Capturing contextual data, or operational information that will contribute in decision making from various internal and external sources of organization.
- ii. **Processing Data:** The captured data is processed into information needed for planning, organizing, coordinating, directing and controlling functionalities at strategic, tactical and operational level.
- iii. **Information Storage:** Information or processed data need to be stored for future use.
- iv. **Retrieval:** The system should be able to retrieve this information from the storage as and when required by various users.
- v. **Information Propagation:** Information or the finished product of the MIS should be circulated to its users periodically using the organizational network.

Characteristics of MIS Following are the characteristics of an MIS:

- i. It should be based on a long-term planning.
- ii. It should provide a holistic view of the dynamics and the structure of the organization.
- iii. It should work as a complete and comprehensive system covering all interconnecting sub-systems within the organization.
- iv. It should be based on need of strategic, tactical and operational information of managers of an organization.
- v. It should also take care of exceptional situations by reporting such situations.
- vi. It should create linkage between all sub-systems within the organization, so that the decision makers can take the right decision based on an integrated view.
- vii. Although the MIS is an integrated, complete system, it should be made in such a flexible way that it could be easily split into smaller sub-systems as and when required.
- viii. A central database is the backbone of a well-built MIS.

What Are The Types Of Information Systems?

1. Knowledge Work System

There are different knowledge management systems that an organization implements to ensure a continuous flow of new and updated knowledge into the company and its processes.

A knowledge work system (KWS) is one of the knowledge management systems that ease the integration of new information or knowledge into the business process. Furthermore, KWS also offers support and resources to various knowledge creation techniques, artificial intelligence applications, and group collaboration systems for knowledge sharing, among others. It also uses graphics, visuals, etc., to disseminate new information.

2. Management Information System

The management information system provides aid to managers by automating different processes that were initially done manually. Business activities like business performance tracking and analysis, making business decisions, making a business plan, and defining workflow. It also provides feedback to the managers by analyzing the roles and responsibilities. A management information system is considered a significant application that helps managers immensely. Here are some of the advantages of the information system:

- i. It enhances the efficiency and productivity of the company.
- ii. It provides a clear picture of the organization's performance.
- iii. It adds value to the existing products, introduces innovation and improves product development.
- iv. It assists in communication and planning for business processes.
- v. It helps the organization provide a competitive advantage

3. Decision Support System

Decision support system is an information system that analyses business data and other information related to the enterprise to offer automation in decision-making or problem-solving. A manager uses it in times of adversities arising during the operation of the business. Generally, the decision support system is used to collect information regarding revenue, sales figures or inventory. It is used across different industries, and the decision support system is a popular information system.

4. Office Automation System

An office automation system is an information system that automates different administrative processes like documenting, recording data, and office transactions, among others. The office automation system is divided into managerial and clerical activities. Here are some of the

business activities that are done under this type of information system: Email, Voice mail, and Word processing

5. Transaction Processing System

The transaction processing system automates the transaction collection, modification, and retrieval process. The peculiar characteristic of this type of information system is that it increases the performance, reliability and consistency of business transactions. It helps businesses perform daily operations smoothly without hassle.

6. Executive Support System

An Executive Support System or ESS helps top-level executives to plan and control workflow and make business decisions. It is very similar to Management Information System or MIS.

7. Expert Systems and Artificial Intelligence These systems use human knowledge captured in a computer to solve problems that ordinarily need human expertise. Mimicking human expertise and intelligence requires that the computer

(1) Recognize, formulate and solve a problem;

(2) Explain solutions and

(3) Learn from experience. These systems explain the logic of their advice to the user; hence, in addition to solving problems they can also serve as a teacher. They use flexible thinking processes and can accommodate new knowledge.

THE IMPORTANCE OF MIS TO BUSINESS ORGANIZATIONS

The importance derived from implementing a management information system includes the following:

- i. It facilitates decision making process of managers. - The routine processing of business transactions is faster and cheaper than manual method.
- ii. It fastens the planning and control functions of management.
- iii. The information contents of MIS can be used by many managers for their varied needs.
- iv. It can be used to achieve competitive advantage in the marketplace.

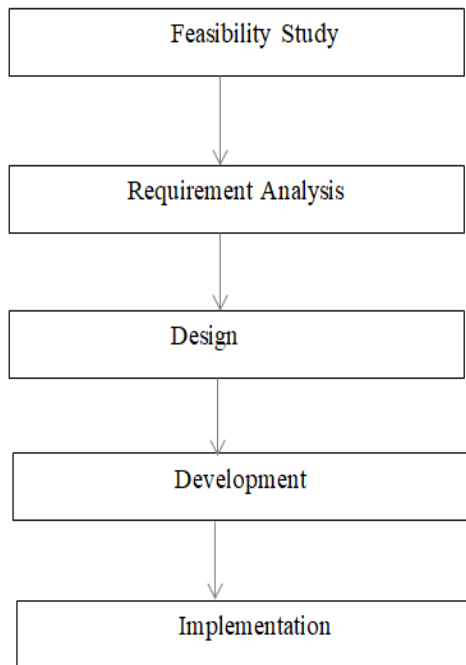
- v. Expected cost savings/reductions - Increase in output volumes and Increase in profit - Increase in accuracy level - Improvement in working environment
- vi. Improvement in turnaround time - Tangible benefits, e.g customer relations and achievement of set goals.
- vii. Provision of improved services to customers.

SYSTEM DEVELOPMENT CYCLE (SDLC)

The systems development life cycle (SDLC) is an approach for designing and developing MIS solutions. It proceeds in stages: feasibility study, requirements analysis, design, development, and implementation. MIS is designed using the systems development life cycle (SDLC). The SDLC is developed in stages. The first stage is to do a feasibility study then the next stage finally.

PHASES OF MIS DEVELOPMENT LIFE CYCLE the five phases are:

- i. Feasibility Study: to know the system is worthwhile, analyze the current situation.
- ii. Requirement Analysis: specifying the requirements that the solution should contain.
- iii. Design: is to design a solution (no programming yet).
- iv. Development: Then the system is developed (programmed) and tested.
- v. Implementation: the system goes live for the end users as it is implemented in the business setting.



DETAILED SDLC

1. Feasibility study: This is the first stage of the systems life cycle. The scope and objectives of the proposed system must be written down. The aim of the feasibility study is to understand the problem and to determine whether it is worth proceeding. There are five main factors to be considered:

Technical feasibility means investigating whether the technology exists to implement the proposed system, or whether this is a practical proposition.

Economic feasibility has to do with establishing the cost-effectiveness of the proposed system if the benefits do not outweigh the costs, then it is not worth going ahead.

Legal feasibility determines whether there is any conflict between the proposed system and legal requirements – for example, will the system contravene the Data Protection Act?

Operational feasibility is concerned with whether the current work practices and procedures are adequate to support the new system. It is also concerned with social factors – how the organizational change will affect the working lives of those affected by the system.

Schedule feasibility looks at how long the system will take to develop, or whether it can be done in a desired time-frame.

The completion of this stage is marked by the production of a feasibility report produced by the systems analyst. If the report concludes that the project should go ahead, and this is agreed by senior managers, detailed requirements analysis will proceed.

2. Requirements analysis: The second phase of systems analysis is a more detailed investigation into the current system and the requirements of the new system. Gathering details about the current system will involve:

- i. Interviewing staff at different levels of the organization from the end-users to senior management.
- ii. Examining current business and systems documents and output. These may include current order documents, computer systems procedures and reports used by operations and senior management.
- iii. Sending out questionnaires and analyzing responses. The questions have to be carefully constructed to elicit unambiguous answers.
- iv. Observation of current procedures, by spending time in various departments. A time and motion study can be carried out to see where procedures could be made more efficient, or to detect where bottlenecks occur.

The systems analysts report will examine how data and information flow around the organization, and may use data flow diagrams to document the flow. It will also establish precisely and in considerable detail exactly what the proposed system will do (as opposed to how it will do it). It will include an in depth analysis of the costs and benefits, and outline the process of system implementation, including the organizational change required. It must establish who the end-users are, what information they should get and in what form and how it will be obtained. Alternative options for the implementation of the project will be suggested. These could include suggestions for:

- i. Whether development should be done in-house or using consultants;
- ii. What hardware configurations could be considered;

- iii. What the software options are. The report will conclude with a recommendation to either proceed or abandon the project.

3. System design: The design specifies the following aspects of a system:

- i. The hardware platform – which type of computer, network capabilities, input, storage and output devices;
- ii. The software – programming language, package or database;
- iii. The outputs – report layouts and screen designs;
- iv. The inputs – documents, screen layouts and validation procedures;
- v. The user interface – how users will interact with the computer system; • The modular design of each program in the application;
- vi. The test plan and test data;
- vii. Conversion plan – how the new system is to be implemented;
- viii. Documentation including systems and operations documentation. Later, a user manual will be produced.

4. System Development This phase includes both the coding and testing of the system, the acquisition of hardware and software.

5. Implementation This phase involves the installation of the new system or conversion of the old system to the new one. The installation phase can include:

- i. Installing the new hardware, which may involve extensive re-cabling and changes in office layouts;
- ii. Training the users on the new system;
- iii. Conversion of master files to the new system, or creation of new master files.

Methods of Conversion There are several different methods of conversion:

- i. **Direct changeover.** The user stops using the old system one day and starts using the new system the next — usually over a weekend or during a slack period. The advantage of this system is that it is fast and efficient, with minimum duplication of work involved. The disadvantage is that normal operations could be seriously disrupted if the new system has errors in it or does not work quite as expected.

- ii. **Parallel conversion.** The old system continues alongside the new system for a few weeks or months. The advantage is that results from the new system can be checked against known results, and if any difficulties occur, operations can continue under the old system while the errors or omissions are sorted out. The disadvantage of parallel conversion is the duplication of effort required to keep both systems running, which may put a strain on personnel.
- iii. **Phased conversion.** This is used with larger systems that can be broken down into individual modules that can be implemented separately at different times. It could also be used where for example only a few customer accounts are processed using the new system, while the rest remain for a time on the old system. Phased conversion could be direct or parallel.
- iv. **Pilot conversion.** This means that the new system will be used first by only a portion of the organization, for example at one branch or factory.

Post-Implementation Review

An important part of the implementation is a review of how the new system is performing, once it has been up and running for a period of time. Minor programming errors may have to be corrected, clerical procedures amended, or modifications made to the design of reports or screen layouts. Often it is only when people start to use a new system that they realize its shortcomings! In some cases they may realize that it would be possible to get even more useful information from the system than they realized, and more programs may be requested. The process of system maintenance, in fact, has already begun, and the life cycle is complete.

System Maintenance

All software systems require maintenance, and in fact the vast majority of programmers are employed to maintain existing programs rather than to write new ones. There are differing reasons for this, and different types of maintenance.

- i. **Perfective maintenance:** This implies that while the system runs satisfactorily, there is still room for improvement. For example, extra management information may be needed so that new report programs have to be written. Database queries may be very slow, and a change in a program may be able to improve response time.

- ii. Adaptive maintenance: All systems will need to adapt to changing needs within a company. As a business expands, for example, there may be a requirement to convert a standalone system to a multiuser system. New and better hardware may become available, and changes to the software may be necessary to take advantage of this. New government legislation may mean that different methods of calculating tax, for example, are required. Competition from other firms may mean that systems have to be upgraded in order to maintain a competitive edge.
- iii. Corrective maintenance: Problems frequently surface after a system has been in use for a short time, however thoroughly it was tested. Some part of the system may not function as expected, or a report might be wrong in some way; totals missing at the bottom, incorrect sequence of data, wrong headings, etc. Frequently errors will be hard to trace, if for example a file appears to have been wrongly updated.

OBJECTIVE OF THE TOTAL SYSTEM

A system can be described as a group of interrelated elements that are integrated with the common purpose of achieving a particular objective by accepting inputs and producing outputs in an organized transformation process. By this definition, an organization such as a firm or a functional area is a system. The organization consists of the physical resources like; Manpower, Material, Machines, Money, and conceptual Information (including data) and they work towards achieving particular objectives that are specified by the owners or management. Other examples of system are biological system, business system, electrical system, data system, education system, operating system, mathematical system, sound system, waste disposal system, computer system, queuing system, etc. The general system theory states that a system is a totality where the whole is greater than the algebraic sum of all the individual elemental parts of the system. Therefore, it is professionally correct for a manager to consider his or her organisation as a system. Such a manager would have made problem solving process easier and more effective.

SUBSYSTEM

A Subsystem is simply a system within a system. This means that systems exist on more than one level that is, Systems can be composed of subsystems or elemental parts. For example, the

system of the firm includes subsystems of the bank may be such departments as savings, demand deposit (checking accounts), and installment loan.

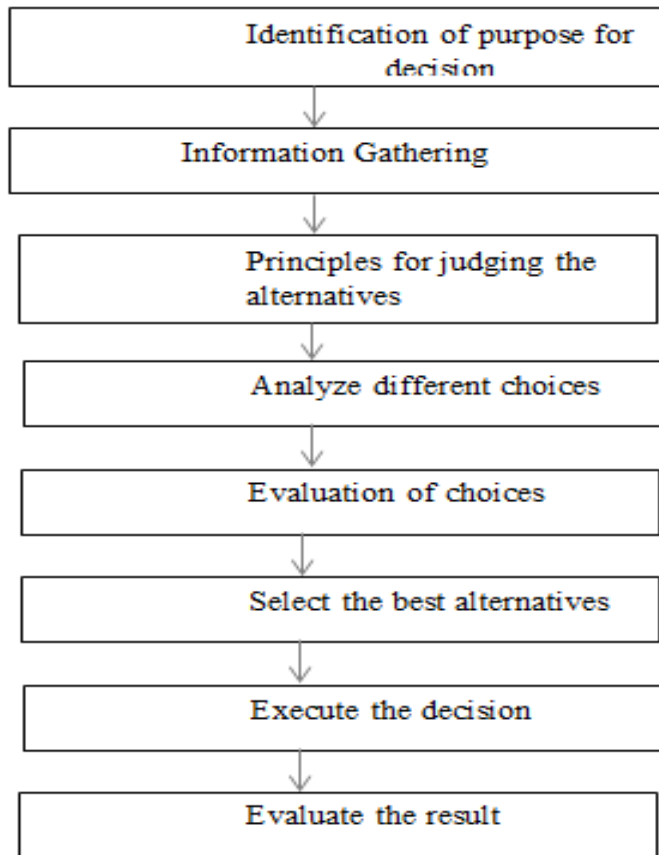
SUPER SYSTEM

When a system is part of a larger system, the larger system is the super system. For example, the system of the firm exists in one or larger environmental systems or super systems. If the firm is a bank, for example, it is part of the financial community. It is also part of the business community, the local community, and the global community. The manager's major responsibility is to ensure that the firm meets its objectives, by ensuring that the various parts of the firm work together as they should.

CONCEPT OF DECISION-MAKING

Decision-making is a cognitive process that results in the selection of a course of action among several alternative scenarios. Decision-making is a daily activity for any human being. There is no exception about that. When it comes to business organizations, decision-making is a habit and a process as well. Effective and successful decisions result in profits, while unsuccessful ones cause losses. Therefore, corporate decision-making is the most critical process in any organization. In a decision-making process, we choose one course of action from a few possible alternatives. In the process of decision-making, we may use many tools, techniques, and perceptions. In addition, we may make our own private decisions or may prefer a collective decision. Usually, decision-making is hard. Majority of corporate decisions involve some level of dissatisfaction or conflict with another party. Let's have a look at the decision-making process in detail.

Decision-Making Process the following are the important steps of the decision-making process. Each step may be supported by different tools and techniques as shown in the figure bellow.



- i. **Step 1:** Identification of the Purpose of the Decision In this step, the problem is thoroughly analyzed. There are a couple of questions one should ask when it comes to identifying the purpose of the decision.
- ii. What exactly is the problem?
- iii. Why the problem should be solved?
- iv. Who are the affected parties of the problem?
- v. Does the problem have a deadline or a specific time-line?

Step 2: Information Gathering A problem of an organization will have many stakeholders. In addition, there can be dozens of factors involved and affected by the problem. In the process of solving the problem, you will have to gather as much as information related to the factors and stakeholders involved in the problem. For the process of information gathering, tools such as 'Check Sheets' can be effectively used.

Step 3: Principles for Judging the Alternatives In this step, the baseline criteria for judging the alternatives should be set up. When it comes to defining the criteria, organizational goals as well as the corporate culture should be taken into consideration. As an example, profit is one of the main concerns in every decision making process. Companies usually do not make decisions that reduce profits, unless it is an exceptional case. Likewise, baseline principles should be identified related to the problem in hand.

Step 4: Brainstorm and Analyze the Choices For this step, brainstorming to list down all the ideas is the best option. Before the idea generation step, it is vital to understand the causes of the problem and prioritization of causes.

For this, you can make use of Cause-and-Effect diagrams and Pareto Chart tool. Cause-and-Effect diagram helps you to identify all possible causes of the problem and Pareto chart helps you to prioritize and identify the causes with the highest effect. Then, you can move on generating all possible solutions (alternatives) for the problem in hand.

Step 5: Evaluation of Alternatives Use your judgment principles and decision-making criteria to evaluate each alternative. In this step, experience and effectiveness of the judgment principles come into play. You need to compare each alternative for their positives and negatives.

Step 6: Select the Best Alternative Once you go through from Step 1 to Step 5, this step is easy. In addition, the selection of the best alternative is an informed decision since you have already followed a methodology to derive and select the best alternative.

Step 7: Execute the decision: Convert your decision into a plan or a sequence of activities. Execute your plan by yourself or with the help of subordinates. **Step 8: Evaluate the Results:** Evaluate the outcome of your decision. See whether there is anything you should learn and then correct in future decision making. This is one of the best practices that will improve your decision-making skills.

DECISION MAKING TECHNIQUES Because they also need to control the timing of all operations, managers set up schedules: They select jobs to be performed during the production process, assign tasks to work groups, set timetables for the completion of tasks, and make sure

that resources will be available when and where they're needed. There are a number of scheduling techniques. We'll focus on two of the most common

Gantt and PERT Charts: Gantt Charts A Gantt chart, named after the designer, Henry Gantt, is an easy-to-use graphical tool that helps operations managers determine the status of projects. Let's say that you're in charge of making the hiking bear that we ordered earlier from the Vermont Teddy Bear Company. Figure 3 "Gantt Chart for Vermont Teddy Bear" is a Gantt chart for the production of one hundred of these bears. As you can see, it shows that several activities must be completed before the bears are dressed: the fur has to be cut, stuffed, and sewn; and the clothes and accessories must be made. Our Gantt chart tells us that by day six, all accessories and clothing have been made. The stuffing and sewing, however (which must be finished before the bears are dressed), isn't scheduled for completion until the end of day eight. As operations manager, you'll have to pay close attention to the progress of the stuffing and sewing operations to ensure that finished products are ready for shipment by their scheduled date as shown in Fig. 3

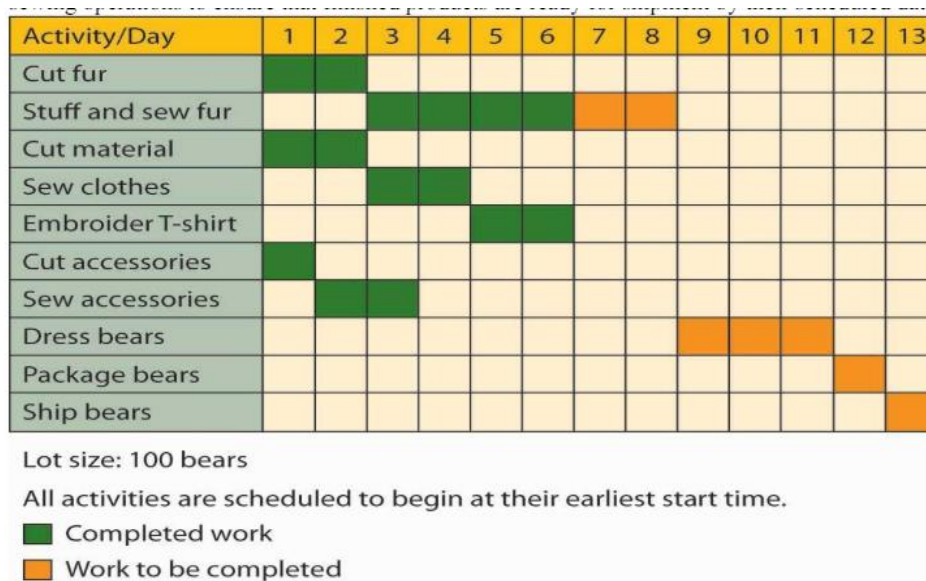


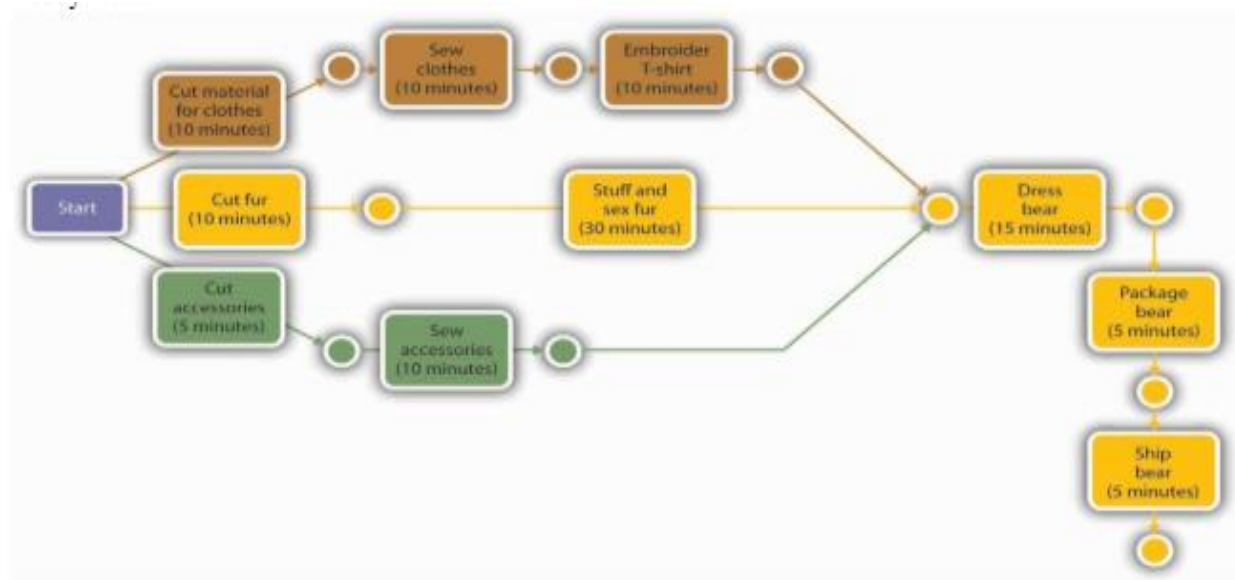
Figure 3: Gantt Chart for Vermont Teddy Bear

PERT Charts Gantt charts are useful when the production process is fairly simple and the activities aren't interrelated. For more complex schedules, operations managers may use PERT charts. PERT (which stands for Program Evaluation and Review Technique) is designed to diagram the activities required to produce a good, specify the time required to perform each

activity in the process, and organize activities in the most efficient sequence. It also identifies a critical path: the sequence of activities that will entail the greatest amount of time. Figure 4

PERT Chart for Vermont Teddy Bear is a PERT diagram showing the same process for producing one —hiker bear at Vermont Teddy Bear.

Figure 4 PERT Chart for Vermont Teddy Bear



Our PERT chart shows how the activities involved in making a single bear are related. It indicates that the production process begins at the cutting station. Next, the fur that's been cut for this particular bear moves first to the stuffing and sewing stations and then to the dressing station. At the same time that its fur is moving through this sequence of steps, the bear's clothes are being cut and sewn and its T-shirt is being embroidered. Its backpack and tent accessories are also being made at the same time. Note that fur, clothes, and accessories all meet at the dressing station, where the bear is dressed and outfitted with its backpack. Finally, the finished bear is packaged and shipped to the customer's house. What was the critical path in this process? The path that took the longest amount of time was the sequence that included cutting, stuffing, dressing, packaging, and shipping a sequence of steps taking sixty-five minutes. If you wanted to produce a bear more quickly, you'd have to save time on this path. Even if you saved the time on

any of the other paths say, the sequence of steps involved in cutting, sewing, and embroidering the bear's clothes you still wouldn't finish the entire job any sooner: the finished clothes would just have to wait for the fur to be stuffed and sewn and moved to the dressing station. In other words, we can gain efficiency only by improving our performance on one or more of the activities along the critical path.

CIPP model The CIPP framework was developed as a means of linking evaluation with programme decision-making. It aims to provide an analytic and rational basis for programme decision-making, based on a cycle of planning, structuring, implementing and reviewing and revising decisions, each examined through a different aspect of evaluation – context, input, process and product evaluation. The CIPP model is an attempt to make evaluation directly relevant to the needs of decision-makers during the phases and activities of a programme. Stufflebeam's context, input, process, and product (CIPP) evaluation model is recommended as a framework to systematically guide the conception, design, implementation, and assessment of service-learning projects, and provide feedback and judgment of the project's effectiveness for continuous improvement. Four aspects of CIPP evaluation these aspects are context, inputs, process, and product. These four aspects of CIPP evaluation assist a decision maker to answer four basic questions:

- i. What should we do? This involves collecting and analysing needs assessment data to determine goals, priorities and objectives. For example, a context evaluation of a literacy program might involve an analysis of the existing objectives of the literacy programme, literacy achievement test scores, staff concerns (general and particular), literacy policies and plans and community concerns, perceptions or attitudes and needs.
- ii. How should we do it? This involves the steps and resources needed to meet the new goals and objectives and might include identifying successful external programs and materials as well as gathering information.
- iii. Are we doing it as planned? This provides decision-makers with information about how well the programme is being implemented. By continuously monitoring the program, decision-makers learn such things as how well it is following the plans and guidelines, conflicts arising, staff support and morale, strengths and weaknesses of materials, delivery and budgeting problems.

- iv. Did the programme work? By measuring the actual outcomes and comparing them to the anticipated outcomes, decision-makers are better able to decide if the program should be continued, modified, or dropped altogether. This is the essence of product evaluation.

MIS REPORT An MIS report is a tool managers use to evaluate business processes and operations. This lesson describes the different kinds of MIS reports and how they are used. It also includes a discussion of how to visually present different kinds of information. What Is an MIS Report? Pretend you are the manager of a medium-sized company's customer service department. Your staff takes phone calls and emails from over 300 customers every day. For the most part, they do a very good job but recently, customers have started to complain that it takes too long to get their questions answered. Upper management at your company is concerned about this and wants to know what they can do to fix the problem. But before they make a decision, they need you to give them more information. How will you do this? This is where MIS reports come in. MIS stands for management information system. Business managers at all levels of an organization, from assistant managers to executives, rely on reports generated from these systems to help them evaluate their business' daily activities or problems that arise, make decisions, and track progress. MIS system reporting is used by businesses of all sizes and in every industry.

Who Uses MIS Reports? MIS systems automatically collect data from various areas within a business. These systems are capable of producing daily reports that can be sent to key members throughout the organization. Most MIS systems can also generate on-demand reports. On-demand MIS reports allow managers and other users of the system to generate an MIS report whenever they need it. Many large businesses have specialized MIS departments, whose only job is to gather business information and create MIS reports. Some of these businesses use sophisticated computing technology and software to gather information. However, the method of collecting information does not have to be that complex. Smaller businesses often use simple software programs and spreadsheets for their MIS reporting needs. There can be as many types of MIS reports as there are divisions within a business. For example, information about sales revenue and business expenses would be useful in MIS reports for finance and accounting managers. Warehouse managers would benefit from MIS reports about product inventory and

shipping information. Total sales from the past year could go into an MIS report for marketing and sales managers.

Type of Information in an MIS Report In our pretend manager example, you've been asked to present information about your department's customer service calls. An MIS report for this would likely contain data such as:

- i. The number of calls your staff takes
 - ii. The number of emails that come in each day
 - iii. The average amount of time it takes to answer a phone call or email
 - iv. The number of questions that your staff answers correctly vs. the number that are incorrect
- To make this information most useful, you also need to ensure that it meets the following criteria: Relevant –

MIS reports need to be specific to the business area they address. This is important because a report that includes unnecessary information might be ignored. **Timely** - Managers need to know what's happening now or in the recent past in order to make decisions about the future. Be careful not to include information that is old. An example of timely information for your report might be customer phone calls and emails going back 12 months from the current date.

Accurate - It's critical that numbers add up and that dates and times are correct. Managers and others who rely on MIS reports can't make sound decisions with information that is wrong. Financial information is often required to be accurate to the dollar. In other cases, it may be OK to round off numbers. **Structured** - Information in an MIS report can be complicated. Making that information easy to follow helps management understand what the report is saying. Try to break long passages of information into more readable blocks or chunks and give these chunks meaningful headings.

GENERAL GUIDELINES FOR FORMATTING FORMS AND REPORTS

- i. A business document that contains some predefined data and may include some areas where additional data are to be filled in. **The Process of Designing Forms and Reports.**
- ii. User-focused activity.
- iii. Follows a prototyping approach.

- iv. Requirements determination:
 - i. Who will use the form or report?
 - ii. What is the purpose of the form or report?
 - iii. When is the report needed or used?
 - iv. Where does the form or report need to be delivered and used?
 - v. How many people need to use or view the form or report? Guidelines for Form and

Report Design

- i. Meaningful titles: clear, specific, version information, current date.
- ii. Meaningful information – include only necessary information, with no need to modify
- iii. Balanced layout: adequate spacing, margins, and clear labels
- iv. Easy navigation system: show how to move forward and backward, and where you are currently

COMMON TYPES OF REPORTS

- i. Scheduled: produced at predefined time intervals for routine information needs
- ii. Key-indicator: provide summary of critical information on regular basis
- iii. Exception: highlights data outside of normal operating ranges
- iv. Drill-down: provide details behind summary of key-indicator or exception reports
- v. Ad-hoc: respond to unplanned requests for non-routine information needs

DATABASE MANAGEMENT SYSTEM (DBMS)

The concept of Database In general, a database system is a computer-based record keeping system i.e. a system whose overall purpose is to record and maintain information and make that information available to user on demand. As a matter of fact, the information concerned can be anything that is deemed to be of significance to the organization system is serving i.e. anything that may be necessary to the decision-making processes involved in the management of that organization. Databases are useful at both the managerial and clerical levels in an organization. At the managerial level are decision-support tools, for reports can be produced from them which summarise what is going on and give pointers for the future. At the clerical level they provide an easy way of entering and retrieving data, and dealing with inquires. So in many situations a clerk

will require access to a database to carry out the latter type of task, and a manager will require access to it to get out a report.

Typical business database applications include:

- i. Stock records
- ii. Personnel records.
- iii. Customer records
- iv. Accounts, and
- v. Mailing lists.

Classification of Database: Databases are classified according to the approaches taken to database organisation. The classes are:-

(a) Relational

(b) Network

(c) Hierarchical Relational database is a sequential representation of the files inform of a table having rows and columns where each row represents a record and the columns corresponding to the fields. The hierarchical database is a tree like representation of the database files. The records of a file are represented in a tree like structure with the fields subordinate to the records in the hierarchy.

The Network database is the representation of the database files. The records relationships are shown by linking.

Features of Database

1. Create and maintain (add, delete, and revise records) a database.
2. Extract and list all records or only those records that meet certain conditions.
3. Make an inquiry.
4. Sort record in ascending or descending order by primary, secondary and tertiary fields.
5. Generated formatted reports with sub-totals and total.

Advantages of database

- i. Reduces the amount of data duplication which occurs using conventional file structures.
- ii. Provides fast and flexible access to information.
- iii. Improves consistency of data stored.
- iv. Provides both data and program independence.
- v. Encourages integration of functional areas in an organisation.
- vi. Avoids duplicating input data to update multiple functionally independent files holding the same data.
- vii. Reduces processing for retrospective changes.

Disadvantages of database

- i. Takes a long time to design and implement.
- ii. Cost more than conventional systems.
- iii. Require high calibre, experienced and specialized personnel.
- iv. Require expensive hardware and software.
- v. Recovery is more difficult in case of damage.

Components of Database System

Database system involves four major components: data, hardware, software, and users. We shall consider each of these briefly below.

- i. Data: - A database is a repository for stored data. In general, it is both integrated and shared. By "integrated" we mean that the database may be thought of as a unification of several otherwise distinct data files, with any redundancy among files partially or wholly eliminated. By "shared" we mean that individual pieces of data in the database may be shared among several different users, in the sense that each of those users may have access to the same piece of data (and may use it for different purposes). Such sharing is really a consequence of the fact that the database is integrated.
- ii. Hardware: - The hardware consists of the secondary storage volumes-disks, drums etc.-on which the data resides, together with the associated devices, control units, channels, and so forth.

- iii. iii. Software: - Database Management System (DBMS) is a collection of software products that is designed to provide a systematic, integrated and flexible approach to organising and accessing data. Between the physical database itself (i.e., the data as actually stored) and the users of the system is a layer of software, usually called the database management system or DBMS. Requests from users for access to the database are handled by the DBMS. One general function provided by the DBases thus the shielding of database users from hardware-level detail. Examples of DBMS include MS Access, FOXPRO, Dbase IV, ORACLE, SQL Server etc.

PARTS OF DBMS SOFTWARE AND SOME TERMINOLOGIES

(i)Data Description Language or Data Definition Language (DDL). Data Description is the process of describing formats and relationships among data elements, as well as how those data elements are to be used. The language used for the data description is called DDL the result of the DDL is called schema. A schema is not the data itself but the description of the data. The schema usually specifies the attributes or characteristics of the data such as: - The data element name -Other names used for the same data element -The type of data -The number of positions - Various integrity rules.

(ii) Data Manipulation Language. The processing of database data is called data manipulation. DML is a set of commands that is used to manipulate data. There are generally two ways to manipulate data in DBMS. -Using a 4GL written database language generally known as query language. -Using a programming language otherwise called data manipulation language (DML) . Both DML and QL statements are embedded in application program at the points where they are needed.

iii.) Report Generator The report generator helps the user to design and generate reports and graphs in printed form. Report headings, column headings, Page numbers, and totals are just some of the features that are easy to include with the report generator. (iv) Data Sublanguage (DSL) The combination of a DDL and DML is often referred to as a DSL. The most common DSL in use today is SQL (Structured Query Language).

(iv)Data Dictionary the DBMS makes use of descriptions of data items provided by DDL. This "data about data" is called a data dictionary. It is a complete description of all data elements in a system.

v.) Users: We consider three broad classes of user. First, there is the application programmer, responsible for writing application programs that use the database. The second class of user, then, is the end-user, accessing the database from a terminal. An end-user may employ a query language provided as an integral part of the system, or he or she may invoke a user-written application program that accepts commands from the terminal and in turn issue requests to the DBMS on the end-user's behalf. The third class of user is the database administrator, or DBA. Database Administrator (DBA):

Responsibilities of DBA

(1) Deciding the Information Content of the Database: It is the DBA's job to decide what information is to be held in a database

(2) Deciding the Storage Structure and Access Strategy: The DBA must also decide how the data is to be represented in the database and must specify their presentation by writing the storage structure definition.

(3) Liaising with Users: It is the business of the DBA to liaise with users, to ensure that the data they require is available.

(4) Defining a Strategy for Backing and Recovering: Once an enterprise is committed to a database system, it becomes critically dependent on the successful operation of that system. In the event of damages to any portion of the database caused by human error or failure in the hardware or supporting operating system, it is essential to be able to repair the data concerned with a minimum of delay and with as little effect as possible on the rest of the system. The DBA must define and implement an appropriate recovery strategy, periodic dumping of the database to a backing tape and provides for reloading from the latest tape.

(5) Monitoring Performance and Responding to Changes in Requirements: The DBA is responsible for so organizing the system as to get the performance that is best for the enterprise and from making the appropriate adjustments as requirements change.

PROJECT MANAGEMENT A project is a well-defined sequence of events with a beginning and an end, directed toward achieving a clear goal, and conducted by people within such established parameters as time, cost, resources, and quality.

A project is different from what you do every day, because a project goal is a specific, non-routine event. Being non-routine, a project requires some planning. How much planning you need depends on the complexity of the project? The more complex the project, the more you have to plan. Planning is necessary to avoid wasting time, resources and effort. For example, if your job is to train employees on a regular basis to use a new spreadsheet application, for you the process requires little project management because the activities are familiar to you. On the other hand, it would be a significant project to develop a completely new course for a new financial reporting system.

Deciding and specifying what to do is the function of project plan. Making sure it is done right is the function of project control. A project is characterized by:

- i.) Must have a definite beginning and a definite ending.
- ii.) Non- repetitive.
- iii.) Utilization of scarce resources.
- vi.) Inability to predict with absolute confidence its final outcome

HOW DOES PROJECT MANAGEMENT DIFFER FROM OTHER MANAGEMENT PRINCIPLES? PROJECT

Management differs in two significant ways as follows:

Firstly, it concentrates on a project with a FINITE life span, whereas departments or other organizational units expect to exist indefinitely. Secondly, projects frequently need resources on a part time basis, whereas permanent organizations try to use resources full time. The sharing of the resources frequently leads to conflict and requires skillful negotiation to see that the projects get the necessary resources to meet objectives throughout their project life.

THE PROJECT LIFE CYCLE

Each project moves through a predictable life cycle of four basic phases with each phase calling for different skills from the project manager. The phases of a project's cycle are:

1. Defining the project;
2. Planning the project;
3. Implementing the plan; and
4. Completing and reviewing the project.

Deciding and specifying what to do is a function of project plan. Making sure that it is done right is the function of project control. Planning is vital if control is required at later stage. The following are main steps in planning a project:

- i. Identify purpose of project
- ii. Set objectives, that is, define deliverables
- iii. Gather information on resources, timescales
- iv. Construct a plan- list tasks, timescales and dependencies, allocate task
- v. Decide on the tools to be used and identify control systems

COMPUTER IN PROJECT MANAGEMENT

The complex nature of modern project makes the use of computers inevitable. The areas of applicability include among others:

- i. Project formulation
- ii. Project feasibility studies
- iii. Appraisal Techniques
- iv. Work Breakdown Structures
- v. Network Analysis
- vi. Costing and Finance
- vii. Material Management
- viii. Monitoring and Evaluation

ix. Report Writing.

Since Project Management involves these three basic phases: Creating a project; Managing a project; and communicating progress, the applicability of computer to such areas enumerated above can be highly appreciated.

WHAT IS MS-PROJECT (Project Management Software Package) MS-Protect is Project management software Package that has capabilities that makes it a necessary tool for project Management. MS-project is a Menu driven program i.e. a program that allows you to interact with the computer by enabling you to select from a group of alternatives, with a highlight of the consequences of selecting any of the various alternatives. Since the essence of project Management is managing information, MS-project's views and features make is easy to enter organize and present project information. MS-project helps plan projects manage and upgrade information and communicate status once a project is underway. For a menu driven package such as MS-project, the use of a mouse device for pointing at things on the screen normally used instead of the keyboard-becomes inevitable.

Type of PROJECT MANAGEMENT TOOLS A project manager has a range of project management tools available to assist with the planning and control of individual systems projects. These tools are designed to improve the effectiveness of the project management process. The project management tools available include:

- i. Work breakdown structure,
- ii. Critical path analysis,
- iii. Gantt chart,
- iv. Resource histogram,
- v. Budget.

Work Breakdown Structure is a tool or method used for breaking down work into manageable units, and then allocating those units to members of the project team. The basic idea of creating a (WBS) is to take the work involved in the project and break it down into smaller and smaller parts, until the project consists of series of work packages. These can then be put into the project plan as activities. Having completed the WBS, it will be possible to construct a project network.

Critical path Analysis (CPA) or Network Analysis: This is a diagram showing the various activities in a project. The aim of the CPA is to identify how those activities link together and to show the critical path or the sequence of activities where delay will result in the overall project. Understanding Critical Tasks: A critical task is one that, if delayed, would also cause the completion of the project to be delayed. Critical tasks are said to be on the critical path. Changes to tasks not on the critical path will not have an effect on the completion date of the project. With Microsoft Project you can quickly identify the critical path, so you can focus on the tasks that require the closest management. Knowing tasks are critical also helps you assess priorities, effectively assign resources, and determine the effect of changes on the project.

Gantt Chart: A Gantt chart, named after the inventor, shows the activities of a project, the same as CPA. However, these activities are presented as a bar chart with the same start and finishing time clearly identified. Resource Histogram: This is a stacked bar chart showing the number and mix of staff required over the duration of a project. Budget: The project budget shows for each month of the project, the proposed expenditure. This will be updated as the project progresses with the actual expenditure and the reason for any variances. Note:- Almost all of these tools can be in some form of computer software, for example, a budget or Gantt chart can be presented on a spreadsheet. Also, Project management software package can produce CPA and Gantt charts from data entered into the software

Computer Information Systems

It refers to managing communications between hardware and software on data storage and management devices. Examples of these devices include databases, cloud storage, and servers. Computer information system architecture depends on the data being collected and how an organization plans to use it. Computer information systems professionals (like systems managers and database administrators) determine the most efficient setup and handle hardware/software integration. Organizational communication involves the relaying of information within the organization from one level to another. Organizations must have a robust environment that encourages and facilitates open communication that, in turn, will lead the employees to accomplish their task effectively.

What is Data?

Data is a collection of raw, unorganized facts and details like text, observations, figures, symbols and descriptions of things etc. In other words, data does not carry any specific purpose and has no significance by itself. Moreover, data is measured in terms of bits and bytes – which are basic units of information in the context of computer storage and processing.

What is Information?

Information is processed, organized and structured data. It provides context for data and enables decision making. For example, a single customer's sale at a restaurant is data – this becomes information when the business is able to identify the most popular or least popular dish.

Difference between Data and Information

Data	Information
Data is unorganized and unrefined facts	Information comprises processed, organized data presented in a meaningful context
Data is an individual unit that contains raw materials which do not carry any specific meaning.	Information is a group of data that collectively carries a logical meaning.
Data doesn't depend on information.	Information depends on data.
Raw data alone is insufficient for decision making	Information is sufficient for decision making
An example of data is a student's test score	The average score of a class is the information derived from the given data.

Summary

What is data capture?

Data capture is the process of extracting information from paper or electronic documents and converting it into data for key systems. It's where most organizations begin their information management and digital transformation journey. Automated data capture solutions reduce

manual data entry and help control the chaos of paper-based processes. Workforces can easily collect relevant data from multiple forms of content and use it for designated workflows.

How does data capture benefit the whole enterprise?

Incorporating these automated capabilities into workflows offers fast, tangible benefits including:

i. Reduces handling of physical paper:

In a remote work setting, digital documents are a must. This method of data processing eliminates the need to transport piles of documents around the office and between buildings, as well as storage, courier and file destruction costs. Paperwork processes become simpler and easier by capturing documents where they originate, on a wide variety of devices, without the need to manually sort or insert separator pages prior to the scanning stage. Staff can use these tools to automatically sort and identify documents, and they can be configured to deliver electronic documents to designated recipients.

ii. Speeds the capture of electronic documents:

Using intelligent capture technology like Brain ware results in an even faster and easier capture process. Intelligent capture: Reads machine-printed characters, hand-printed characters, barcodes and marks like check boxes and filled circles

Captures all file formats (like Word, Excel and images) automatically from email inboxes and file-sharing services

Imports documents directly from most applications, further streamlining workflows

iii. Eliminates manual data entry:

Organizations reduce errors as well as the most tedious aspects of paperwork processing. That means employees won't have to manually process business checks, or go through mail to extract information.

Even for a small team, reducing the manual data entry effort surrounding document processing lowers labor costs, accelerates business processes and improves customer service by ensuring

employees have up-to-date information at their fingertips. Scaling these capabilities across a larger department or organization yields even better results.

Input device

An **input device** is any hardware device that sends data to a computer, allowing you to interact with and control it. The picture shows a Logitech trackball mouse, an example of an input device. The keyboard and mouse are a computer's most commonly used or primary input devices. However, other devices also input data into a computer.

Types of input devices

- i. Audio conversion device
- ii. Barcode reader
- iii. Biometrics (e.g., fingerprint scanner).
- iv. Business card reader

Data collection device

- i. Digital Camera, and digital camcorder.
- ii. EEG (Electro Encephalo Graphy),
- iii. Finger (with touch screen or Windows Touch).
- iv. Gamepad, balance board, dance pad, joystick, paddle, steering wheel, pedals, and Microsoft Kinect.
- v. Gesture recognition
- vi. Graphics tablet
- vii. Keyboard
- viii. Light gun
- ix. Light pen
- x. Magnetic ink (like the ink found on checks).
- xi. Magnetic stripe reader
- xii. Touch-sensitive pad (like pad on early iPod)
- xiii. Touch screen
- xiv. Voice (using voice speech recognition or biometric verification).

- xv. Video capture device
- xvi. Virtual device
- xvii. Webcam

What is data control?

Data control involves the collection, storage and use of consumer information by companies.

To put it into practice, organizations must leverage modern and secure tools that not only facilitate data gathering, but also ensure its utmost protection, mitigating the risk of any potential breaches. When customer information is lost or mishandled, it becomes challenging to understand their consumption habits and identify the most effective communication channels, directly impacting the accuracy of managerial decision-making. To recognize the importance of information as the foundation for strategic decisions; some companies now employ Artificial Intelligence (AI) to enhance data management efficiency.

What is a Management Information System?

An MIS is a system that provides managers with the necessary information to make decisions about an organization's operations. The MIS gathers data from various sources and processes it to provide information tailored to the managers' and their staff's needs.

While businesses use different types of systems, they all share one common goal: to provide managers with the information to make better decisions. In today's fast-paced business environment, having access to accurate and timely information is critical for success. MIS allows managers to track performance indicators, identify trends, and make informed decisions about where to allocate resources.

Importance of management information systems for businesses

MIS allow businesses to have access to accurate data and powerful analytical tools to identify problems and opportunities quickly and make decisions accordingly.

A management information system should do the following:

- i. Provide you with information you need to make decisions.

- ii. Can give you a competitive edge by providing timely, accurate information.
- iii. Can help you improve operational efficiency and productivity
- iv. Allows you to keep track of customer activity and preferences.
- v. Enables you to develop targeted marketing campaigns and improve customer service

Decision Making

Decision making is the process of choosing one course of action from a set of options. Most decision-makers are taking in information, examining and selecting options, predicting likely outcomes based on those options, considering the relative value of the options, all in light of particular goals. Within organizations, it is also important to consider how often certain kinds of decisions must be made and what kind of information is available to use in making the decision.

Organizational Structure

Business organizational structures came about to simplify decision-making. The typical organization chart depicts a hierarchy of authority starting with the chairman, CEO and other executive officers at the top because the executive officers set the goals and direction of the company. Supporting layers of managers under the executive level are tasked with applying these goals to their areas of responsibility or departments, which are arranged in order of appropriate information flow from management.

Information flow

Information flow in an organization refers to the process of transmitting and receiving information throughout different parts of the organization. It is a complex process that involves sharing and exchanging information between various departments and individuals. Efficient information flow is crucial for organizational communication, innovation, and decision-making. It helps in generating and disseminating information within the organization, which can impact organizational climate, morale, and overall effectiveness. Information flow is facilitated by technologies and systems that connect material flows with decision-makers.

Transaction Processing System (TPS) is a type of system which usually helps an organization to handle its daily business operations and transactions in an efficient and effective manner by processing and recording each operation. A transaction is an occurrence in which goods,

services, or money are passed from one person, account, etc., to another. Such operations/transactions may include customer orders, purchase orders as well as handling of invoices. A TPS is therefore responsible for collecting, storing, modifying and retrieving data pertaining to the transactions that have taken place in an organization and finally generate reports which are used by other levels of management. The characteristics of a TPS include performance, reliability and consistency. Transaction Processing Systems are usually used at the operational level (by employees who are at the bottom level) of an organization's hierarchy.

An example of a TSS is a point of sale in a supermarket such as Nakumatt which is used to record each sale transaction which takes place in the supermarket. At the end of the day, a record of all transactions can be generated from the TPS in order to view which products have been sold.

Management Information System (MIS) is a system which brings together people (the management), information as well as the systems (both hardware and software). This type of system is important in an organization as it provides information that is essential to operations, management and decision making functions. Some of these functions include planning, controlling, decision making, organizing, and staffing. Management Information Systems are usually used at the tactical level (by employees who are at the middle level) of the organization's management hierarchy as shown on Figure 2 below. An example of an MIS is the University of Nairobi's Student Management Information System (SMIS) which can be used to generate reports about the registration status of the students in order to determine who is eligible to sit for the end of semester examinations.

Result Tables

A result table represents a collection of tabular data that resides neither in a database nor in program memory. Instead, it is generated as the result of a database query.

NOTE: A result table of the DB Interface Module corresponds to the terms *table expression*, *select expression*, *result set*, and *record set*, which are used in other languages.

Result tables are produced by the `table ()` method of class `RWDBResult`. Class `RWDBResult` represents a sequence of zero or more `RWDBTable` instances.

Decision tables are a concise visual representation for specifying which actions to perform depending on given conditions. Decision table is the term used for a Control table or State-transition table in the field of Business process modeling; they are usually formatted as the transpose of the way they are formatted in Software engineering.

A decision table is considered *balanced* or *complete* if it includes every possible combination of input variables. In other words, balanced decision tables prescribe an action in every situation where the input variables are provided.

The following **balanced decision table** is an example in which a technical support company writes a decision table to enable technical support employees to efficiently diagnose printer problems based upon symptoms described to them over the phone from their clients.

Printer troubleshooter									
Conditions	Printer prints	Rules							
		No	No	No	No	Yes	Yes	Yes	Yes
		Yes	Yes	No	No	Yes	Yes	No	No
		No	Yes	No	Yes	No	Yes	No	Yes
Actions	Check the power cable			✓					—
	Check the printer-computer cable	✓		✓					—
	Ensure printer software is installed	✓		✓		✓		✓	—
	Check/replace ink	✓	✓				✓		—
	Check for paper jam		✓		✓				—

This is just a simple example, and it does not necessarily correspond to the reality of printer troubleshooting. Even so, it demonstrates how decision tables can scale to several conditions with many possibilities.